

OpenAI and Navier-Stokes

Evidence review / September 2026

This is an offline demonstration report prepared for an Aether film. It is not a record of a live literature search or a newly verified scientific result. The distinction between numerical progress and a proof is the central subject of this report.

What would count as a solution?

The scope of the Millennium Prize Problem

The official problem asks for existence and smoothness in three dimensions, or a valid breakdown example, under its stated conditions. A claim must specify the domain, regularity class, initial data, forcing, and the exact conclusion. Numerical success on a finite grid does not supply a general proof.

Three different kinds of progress

Computation / restricted theorems / a general proof

A numerical solver may improve prediction at finite resolution. A restricted theorem may establish a result for small data, a special geometry, or extra hypotheses. A general solution must address the official three-dimensional statement. These results must not be conflated, even when the same mathematical equations appear in all three.

How to audit a headline

Trace the statement back to its evidence

Locate the original announcement and full paper. Record the version and date.

Compare the paper's hypotheses with the official problem. Inspect whether the key step is proved, assumed, numerically observed, or delegated to a model. Seek independent expert assessment. Repeated reporting of a single claim is not independent verification.

Evidence matrix

A practical standard for interpretation

Claim: general 3D regularity. Required evidence: a complete argument under the official hypotheses, available for independent scrutiny. Claim: numerical benchmark improvement. Required evidence: data, baselines, resolution, boundary conditions, error measures, and replication. Claim: AI-assisted proof discovery. Required evidence: the final mathematical proof, independent of how it was discovered.

Conclusion and reference entry points

What can responsibly be said

Do not describe an AI fluid benchmark or special-case theorem as a solution to the Millennium Prize Problem. Until a specific complete claim has been independently assessed, its status remains unverified. Reference entry points: Clay Mathematics Institute, <https://www.claymath.org/millennium/navier-stokes-equation/> ; arXiv, <https://arxiv.org/search/?query=navier+stokes&searchtype=all> ; OpenAI research, <https://openai.com/research/> . These are reference entry points, not citations to a verified OpenAI solution.